

Module No : Module – 03
Assignment No: Assignment 03
Module Title: Software Testing Life Cycle with AI

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Q1.Create a Test Plan for an Online Shopping Website.

Answer: A Test Plan is a document that describes the testing scope, objectives, resources, schedule, test approach and responsibilities for a software application.

Test Plan – Online Shopping Website

1. Objective – The objective of testing is to verify that the Online Shopping Website works correctly and meets the required business and user requirements. The testing will ensure that users can register, log in, search products, add products to the cart, check out, and place orders successfully.

2. Scope

In Scope –

- User Registration
- User Login/Logout
- Product Search
- Product Details
- Add to cart
- Update/ Remove cart Items
- Checkout
- Payment
- Order Confirmation
- Order history

Out of Scope

- Testing external systems that are not controlled by the website team.

3. Testing Types – The following types of testing will be performed:

- Functional Testing
- UI Testing
- Usability Testing
- Compatibility Testing
- Performance Testing
- Security Testing

4. Test Environment – Testing will be performed using

- Operating System: Windows / Android
- Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge
- Devices: Desktop, Laptop, Mobile

- Internet: Stable internet connection
- Test Server: QA/Staging environment

5. Test Data – The following test data will be used:

- Valid and invalid email addresses
- Valid and invalid passwords
- New and existing user accounts
- Product names and product IDs
- Different product quantities
- Valid and invalid shipping addresses
- Valid and invalid coupon codes
- Valid and invalid payment information

6. Testing Activities – The QA team will perform the following activities:

- Review the requirements.
- Prepare the Test Plan.
- Create test cases and checklists.
- Prepare test data.
- Set up the test environment.
- Execute test cases.
- Identify and report defects.
- Retest fixed defects.
- Perform regression testing.
- Prepare the final test report.

7. Entry Criteria – Testing will start when:

- Requirements are finalized and available.
- The application is deployed to the test environment.
- The test environment is ready.
- Required test data is available.
- Test cases are prepared and reviewed.
- Major build-blocking issues are resolved.

8. Exit Criteria – Testing will be completed when:

- All planned test cases have been executed.
- Critical and high-priority defects are fixed and retested.

- Required regression testing is completed.
- No major business-blocking defects remain.
- Test results and final reports are prepared.

9. Deliverables – The following testing documents will be prepared:

- Test Plan
- Test Scenarios
- Test Cases
- Test Checklist
- Test Data
- Defect/Bug Reports
- Test Execution Report
- Final Test Summary Report

10. Risks – Possible risks include:

- Website may become unavailable during testing.
- Test environment may not work properly.
- Payment testing may fail because of third-party services.
- Requirements may change during testing.
- Some bugs may not be fixed on time.
- Different browsers or devices may show different problems.

This Test Plan provides a simple approach for testing an online shopping website. It helps ensure that the website is functional, user-friendly, secure, and ready for customers.

Q2.Prepare a Test Checklist. Create a checklist containing at least 20 validation points for the below features. Login Checklist, Registration Checklist, Checkout Checklist.

Answer: Table for Login Checklist.

Test Area	Validation	Expected Result
Login	Enter valid email and valid password	User should log in successfully
Login	Enter valid email and wrong password	Login should fail and an error message should appear
Login	Enter unregistered email	System should show an appropriate error message.
Login	Leave email field empty	System should ask the user to enter an email.
Login	Leave password field empty	System should ask the user to enter a password
Login	Leave both fields empty	Required-field validation messages should appear.
Login	Enter an invalid email format	System should show an invalid email message.
Login	Check password field	Password should be hidden/masked.
Login	Click “Forgot Password”	User should be taken to the password recovery process.
Login	Logout after successful login	User should be logged out successfully.

Table for Registration Checklist

Test Area	Validation	Expected Result
Registration	Enter all valid registration information	Account should be created successfully.
Registration	Enter an already registered email	System should show required-field messages.
Registration	Enter invalid email format	System should reject the email and show an error.
Registration	Leave required fields empty	System should show required-field messages.
Registration	Enter a weak password	System should show password requirements or reject the password
Registration	Enter a valid phone number	Phone number should be accepted
Registration	Try registering without accepting required terms	Register should not continue
Registration	Enters letters in a phone number field	System should reject invalid characters.

Table for Checkout Checklist

Test Area	Validation	Expected Result
Checkout	Checkout with a valid product in cart	User should be able to proceed to checkout
Checkout	Checkout with an empty cart	System should prevent checkout and show an appropriate message.
Checkout	Enter valid shopping address	Address should be accepted successfully.
Checkout	Leave required shipping fields empty	System should show required-field validation message.
Checkout	Select a valid payment method	Selected payment method should be accepted
Checkout	Verify subtotal and total calculation	Amounts should be calculated correctly.
Checkout	Try to place an order with missing required information	Order should not be placed until required information is provided.
Checkout	Place an order with all valid information	Order should be placed successfully and confirmation should be displayed.
Checkout	Check order total before payment	Total should correctly include product price, quantity, shipping cost, discount, and applicable taxes.
Checkout	Increase/decrease product quantity	Cart total should update correctly.
Checkout	Enter an invalid/expired coupon code	System should reject the coupon and show an appropriate message.